Abraham Noumsi

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Education

Institut Universitaire Saint Jean du Cameroun — Eyang, Cameroon
Engineering degree in Computer Science and Information Systems — *September 2023 – October 2028 (Expected)*

Professional Experience

Python/Odoo Developer Intern — June 2025 – September 2025
N2T SOLUTIONS SARL — *Yaounde, Cameroon*

- Executed the technical ERP migration for the Port Authority of Kribi from Odoo 8 to 17, delivering **zero operational downtime** for **500+ daily port operators**.
- Engineered backend modules in **Python** and **PostgreSQL**, refactoring legacy business logic to boost transactional processing throughput by **30%**.
- Built automated ETL and validation pipelines migrating **100,000+ historical records** with **100% schema integrity** verified through regression test suites.

IT Systems Intern — June 2024 – August 2024
SEJOCE SARL — *Douala, Cameroon*

- Maintained **99.9% uptime** across mission-critical retail and warehouse servers by diagnosing network bottlenecks and scheduling proactive maintenance.
- Digitized inventory and distribution tracking records, enforcing database-level constraints to eliminate stock audit discrepancies.
- Resolved technical hardware and network incidents for **50+ on-site workstations**, establishing standard operating procedures to reduce ticket resolution time.

Personal Projects

Umzugsflink | *Next.js, TypeScript, Medusa.js, PostgreSQL, Stripe, Redis*

- Architected a multi-vendor moving marketplace (**Turborepo monorepo**) pairing clients with student movers, featuring a high-concurrency **Next.js 15 App Router** storefront and headless **Medusa.js v2** backend.
- Implemented split-escrow payments via **Stripe Connect**, holding client funds in escrow and automating contractor payouts upon verified job completion with **0% transaction discrepancy**.
- Integrated real-time negotiation and dispatch using **TalkJS** with dynamic quote adjustments, booking scheduling, and asynchronous **Redis webhook queues**.

Bwaka | *React Native, TypeScript, Supabase, PostgreSQL*

- Developed a nationwide lost-and-found mobile app in **React Native**, implementing carrier-prefix authentication (**MTN, Orange, Camtel**) with **automated SMS OTP** via Supabase Auth.
- Engineered an automated matching engine using **PostgreSQL trigram indexing (pg_trgm)**, correlating lost and found reports across Cameroon's **10 regions** with **under 500ms query latency**.
- Implemented client-side **OCR with automatic PII masking** on identity documents (CNI, passports) and challenge-question gating, preventing fraudulent ownership claims.

Orvia | *NestJS, TypeScript, Angular, React Native, Prisma, RAG*

- Architected an enterprise document intelligence pipeline with **Retrieval-Augmented Generation (RAG)**, integrating **spatial bounding-box OCR** to ground all AI answers with **page-exact source citations**.
- Built an **Angular** enterprise portal featuring multi-tenant isolation, granular **RBAC**, and an immutable **SHA-256 hash-chained audit trail** ensuring verifiable compliance.
- Delivered an offline-first **React Native mobile client** with bidirectional **SQLite delta-synchronization**, enabling local queries across indexed document collections without network access.

KCS Media Studio | *Python, Whisper AI, FastAPI, React, PyTorch, FFmpeg*

- Fine-tuned **OpenAI Whisper** acoustic models and developed NLP translation pipelines tailored for Cameroonian local languages (**Ngomala, Ewondo, Bassa**) with **over 85% transcription accuracy**.
- Engineered an interactive waveform timeline editor supporting **millisecond-accurate caption adjustments**, speaker diarization, and accelerated video subtitle burning via **FFmpeg**.
- Built an asynchronous backend with **FastAPI and WebSockets**, streaming real-time transcription status, chunked audio processing, and live alignment states under concurrent client loads.

AKIBA | *Next.js, Express.js, TypeScript, PostgreSQL, React Native*

- Architected a full-stack home services marketplace connecting consumers across Cameroon with **200+ vetted artisans and technicians**.
- Built a low-latency geo-matching search engine using **PostgreSQL spatial indexing**, reducing service discovery time and driving a **40% increase in booking conversions**.
- Shipped unified web and **React Native mobile applications** sharing domain models and validation schemas, ensuring seamless booking and review flows.

We Choose! | *Go, TypeScript, PostgreSQL, Docker*

- Developed a tamper-evident digital voting platform deployed via **Docker**, securely administering university student government elections for **3,000+ registered student voters**.
- Implemented a **SHA-256 cryptographic ballot-hashing protocol** guaranteeing voter anonymity and auditability while mathematically preventing double-voting.
- Automated candidate verification and **sub-second ballot tallying** under peak concurrent traffic spikes, cutting election administration overhead by **60%**.

Chaxiraxi (Gibbon Core) | *PHP, MySQL, TypeScript, Tailwind CSS*

- Led the frontend modernization and component design system (**"Chaxiraxi"**) for **Gibbon Core**, an open-source student information platform using **PHP, MySQL, and Tailwind CSS**.
- Replaced legacy monolithic server-rendered markup with accessible, responsive modular components, reducing page load latency by **35%** across administrative modules.
- Built an instant keyboard-driven search indexer (**fast-finder**) and standardized data-entry workflows across student records and academic grade management.

Technical Skills

Languages: Python, Go, Java, TypeScript, JavaScript, SQL, R, C, PHP, HTML/CSS
Backend & Distributed Systems: NestJS, Node.js, FastAPI, Spring Boot, RESTful APIs, gRPC, WebSockets, Odoo ORM
Frontend, Mobile & CMS: Next.js, React, React Native, Angular, Tailwind CSS, WordPress, Turborepo
Data Analysis & Databases: PostgreSQL (pgvector), Supabase, Redis, SQLite, Pandas, NumPy, Data Modeling, ETL Pipelines
AI Engineering & Media: Retrieval-Augmented Generation (RAG), LangChain, Whisper STT, PyTorch, Hugging Face, OCR, FFmpeg
DevOps & Monitoring: Docker, Linux, CI/CD (GitHub Actions), Monitoring (Grafana, Sentry), Git, Postman, Stripe Connect
Design & Prototyping: Figma, Adobe Creative Suite (Photoshop, Illustrator), UI/UX Prototyping

Honors & Awards

- **Hult Prize OnCampus IUSJC 2025 | Campus Champion Team**, selected among the **top 6 national finalist teams** in Cameroon for social venture innovation.

Leadership & Community Engagement

Student Ambassador 2024 – Present
Institut Universitaire Saint Jean du Cameroun *Yaounde, Cameroon*

- Promoted engineering education and mentored student cohorts on hands-on IoT hardware and software projects.

Organizing Committee Member – Hult Prize IUSJC 2026 – Present
Hult Prize Foundation *Yaounde, Cameroon*

- Drove campus event marketing, participant onboarding, and pitch training for competing student entrepreneurship teams.

Professional Skills, Languages & Interests

Soft Skills: Cross-functional teamwork, Persuasive communication, Analytical problem-solving, Initiative, Autonomy
Languages: **French** (Native speaker), **English** (Professional working proficiency / B2 – Technical documentation, professional communication), **Spanish** (Elementary proficiency / Basic conversational notions)
Interests: Chess, Origami, Reading